Problem 8 | Problems T6andT7_Electric and Magnetic Properties.pdf #8 | Intrinsic carriers · electrons and holes

Carrier Concentrations of Intrinsic Silicon from its Conductivity

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8. The electrical conductivity of intrinsic Si at room temperature is $4 \cdot 10^{-4}\ \Omega^{-1}m^{-1}$*, and the mobilities of electrons and holes are 0.14 y 0.048 m²/(V·s) respectively. Calculate the concentration of holes and electrons.*

The statement is transcribed as printed: "y" is the Spanish "and".

Official Solution: 1.32×10^{16} m⁻³

HYPOTHESES & PARAMETERS

Hypotheses

1. **Two carrier types.** In an intrinsic semiconductor each excitation creates one electron in the conduction band and one hole in the valence band, so both carry current: $\sigma = n|q_e|\mu_e + p|q_e|\mu_h$ [Session 13 Slides 27-28].
2. **Equal numbers.** Because the carriers are created in pairs, $n = p = n_i$ and the charge of the hole has the same magnitude as that of the electron, $|q_h| = |q_e|$ [Session 13 Slide 28].
3. **Room temperature,** all values are those of the statement. The charge is 1.602×10^{-19} C (the value that reproduces the key).

Data

Quantity	Symbol	Value	In SI
Conductivity	σ	4×10^{-4} Ω⁻¹m⁻¹	same
Electron mobility	μ_e	0.14 m²/(V·s)	same
Hole mobility	μ_h	0.048 m²/(V·s)	same
Charge	$\lvert q_e \rvert$	1.602×10^{-19} C	same

Unknown: the hole concentration p and the electron concentration n.

MODEL & STRATEGY

The two carriers add their currents. As their numbers are equal, the conductivity is the carrier concentration times the charge times the sum of the two mobilities, and that one equation has one unknown. The figure shows the excitation of pairs and the way the conductivity splits between the carriers.

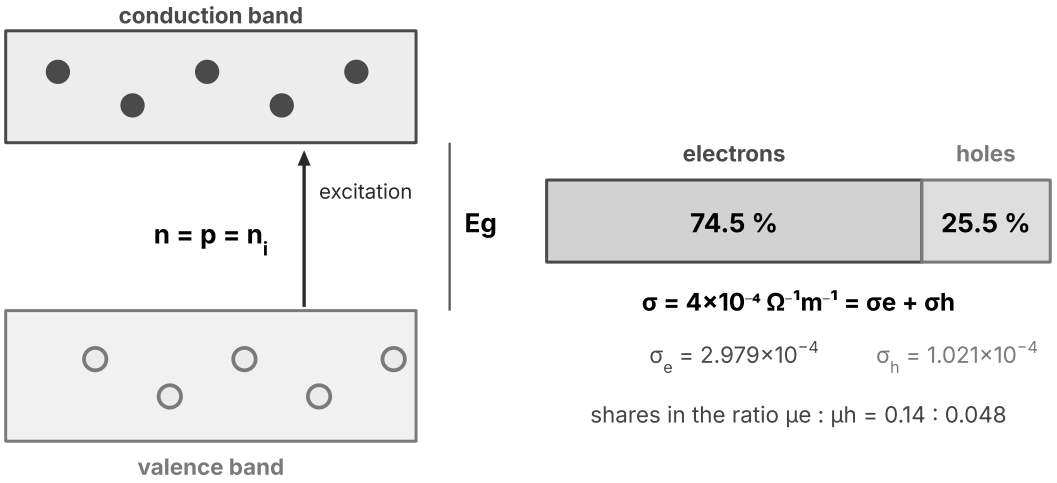


Figure 8. Problem 8: an intrinsic semiconductor has as many electrons as holes (n = p = ni, the intrinsic concentration); the conductivity splits in the ratio of the mobilities, 0.14 to 0.048 (schematic band diagram, bar to scale).

Strategy.

1. Write the conductivity with both carriers and impose $n = p$.
2. Solve for the concentration.
3. Evaluate with SI values, then compute the share of each carrier.
4. Compare with the intrinsic concentration of silicon tabulated on Slide 31.

STEP-BY-STEP CALCULATION

1. Conductivity with both carriers [Session 13 Slides 27-28]

$$\sigma = n\,|q_e|\,\mu_e + p\,|q_e|\,\mu_h$$

2. Condition of pairs and solution for the concentration

Substitution of $n = p = n_i$ and extraction of the common factor:

$$\sigma = n_i \left|q_e\right| \mu_e + n_i \left|q_e\right| \mu_h = n_i \left|q_e\right| \left(\mu_e + \mu_h\right)$$

Division by $|q_e|(\mu_e + \mu_h)$:

$$n_i = \frac{\sigma}{\left|q_e\right| \left(\mu_e + \mu_h\right)}$$

3. Numerical evaluation, one operation per block

Sum of the mobilities:

$$\mu_e + \mu_h = 0.14 \ \frac{\mathrm{m}^2}{\mathrm{V\,s}} + 0.048 \ \frac{\mathrm{m}^2}{\mathrm{V\,s}} = 0.188 \ \frac{\mathrm{m}^2}{\mathrm{V\,s}}$$

Denominator:

$$\left|q_e\right| \left(\mu_e + \mu_h\right) = 1.602 \times 10^{-19} \ \mathrm{C} \times 0.188 \ \frac{\mathrm{m}^2}{\mathrm{V\,s}} = 3.01176 \times 10^{-20} \ \frac{\mathrm{C\,m}^2}{\mathrm{V\,s}}$$

Quotient:

$$n_i = \frac{4 \times 10^{-4} \ \Omega^{-1}\mathrm{m}^{-1}}{3.01176 \times 10^{-20} \ \mathrm{C\,m^2 V^{-1} s^{-1}}} = 1.3281 \times 10^{16} \ \frac{\Omega^{-1}\mathrm{m}^{-1}\ \mathrm{V\,s}}{\mathrm{C\,m}^2}$$

Reduction of the units, with $\Omega^{-1} = \mathrm{A/V}$ and $\mathrm{A\,s} = \mathrm{C}$:

$$\frac{\Omega^{-1}\mathrm{m}^{-1}\,\mathrm{V\,s}}{\mathrm{C\,m}^2} = \frac{\mathrm{A\,V\,s}}{\mathrm{V\,C\,m}^3} = \frac{\mathrm{C}}{\mathrm{C\,m}^3} = \mathrm{m}^{-3}$$

$$\mathbf{n = p = n_i = 1.328 \times 10^{16}\ m^{-3}}$$

4. Share of each carrier in the conductivity

Electron and hole conductivities:

$$\sigma_e = n\,|q_e|\,\mu_e = 1.3281 \times 10^{16} \times 1.602 \times 10^{-19} \times 0.14\ \Omega^{-1}\mathrm{m}^{-1} = 2.979 \times 10^{-4}\ \Omega^{-1}\mathrm{m}^{-1}$$

$$\sigma_h = p\,|q_e|\,\mu_h = 1.3281 \times 10^{16} \times 1.602 \times 10^{-19} \times 0.048\ \Omega^{-1}\mathrm{m}^{-1} = 1.021 \times 10^{-4}\ \Omega^{-1}\mathrm{m}^{-1}$$

Shares (they equal the mobility ratios because $n = p$):

$$\frac{\sigma_e}{\sigma} = \frac{0.14}{0.188} = 0.7447 \qquad \frac{\sigma_h}{\sigma} = \frac{0.048}{0.188} = 0.2553$$

Sum:

$$\sigma_e + \sigma_h = 2.979 \times 10^{-4} + 1.021 \times 10^{-4} = 4.000 \times 10^{-4}\ \Omega^{-1}\mathrm{m}^{-1}$$

5. Comparison with the table of Slide 31

Slide 31 gives for silicon $n_i = 1.5 \times 10^{16}$ m⁻³, $\mu_e = 0.135$ and $\mu_h = 0.048$ m²/(V·s):

$$\sigma = 1.5 \times 10^{16} \times 1.602 \times 10^{-19} \times (0.135 + 0.048)\ \Omega^{-1}\mathrm{m}^{-1} = 4.40 \times 10^{-4}\ \Omega^{-1}\mathrm{m}^{-1}$$

$$\frac{1.328 \times 10^{16}}{1.5 \times 10^{16}} = 0.885$$

RESULT & CHECKS

Checks

- **Units.** $\sigma/(|q|\mu)$ has units of m^{-3} (step 3).
- **Reconstruction.** The two partial conductivities add up to the given 4×10^{-4} $\Omega^{-1}\mathrm{m}^{-1}$ (step 4).
- **Order of magnitude.** Relative difference of the computed concentration from the tabulated 1.5×10^{16} m^{-3}; it comes from the rounded conductivity of the statement (4×10^{-4}, with $\mu_e = 0.14$) against the 4.4×10^{-4} that the table values give:

$$\frac{1.328 \times 10^{16} - 1.5 \times 10^{16}}{1.5 \times 10^{16}} = -0.115$$

- **Concentration against atom density.** With $N_{\mathrm{Si}} = 4.995 \times 10^{28}$ atoms/m³ (from $\rho_m N_A / M$, Problem T6-09):

$$\frac{N_{\mathrm{Si}}}{n_i} = \frac{4.995 \times 10^{28}\ \mathrm{m}^{-3}}{1.328 \times 10^{16}\ \mathrm{m}^{-3}} = 3.76 \times 10^{12}$$

one carrier pair per 3.8×10^{12} atoms, which is why intrinsic silicon is such a poor conductor.

Interpretation

Silicon at room temperature has 1.3×10^{16} electrons and the same number of holes per cubic metre, and three quarters of the current is carried by the electrons because they move almost three times faster. If the holes had the mobility of the electrons the two shares would be equal; if the number of electrons exceeded that of the holes (doping) the material would no longer be intrinsic.

Check against the official solution

Item	Official	Computed	Comment
Concentration of holes and electrons	1.32×10^{16} m^{-3}	1.328×10^{16} m^{-3} for each	agrees to 0.6 %, see the warning

DISCREPANCY WITH THE OFFICIAL SOLUTION

The key prints 1.32×10^{16} m^{-3}. Our exact value is 1.3281×10^{16} m^{-3}, which rounds to 1.33×10^{16}: the key appears to truncate instead of rounding. With the rounded charge 1.6×10^{-19} C the result would be 1.330×10^{16} m^{-3}. We keep the computed value.